

VESYOLAYA GORKA, Russian Federation

WMO: 314180

Lat: 52.283N

Lon: 135.800E

Elev: 201

StdP: 98.93

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.6	-30.9	-36.5	0.1	-31.1	-34.5	0.1	-29.9	6.4	-12.3	5.4	-10.5	0.2	20	0.571

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.5	29.3	20.3	27.2	19.3	25.4	18.4	22.1	27.2	20.8	25.4	19.6	23.7	1.5	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.4	15.4	24.8	19.1	14.2	23.2	18.0	13.3	22.1	66.0	27.3	61.2	25.5	57.1	23.7	33.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.3	4.1	3.2	-34.2	32.7	2.3	2.2	-35.9	34.3	-37.3	35.5	-38.6	36.7	-40.3	38.3		
(5)			-34.4	24.5	2.3	2.4	-36.0	26.2	-37.3	27.6	-38.6	28.9	-40.3	30.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.1	-21.7	-17.3	-8.3	1.7	9.2	15.4	18.7	17.5	11.7	2.6	-10.6	-20.8	(6)
(7)		DBStd	15.25	5.75	4.97	5.06	4.53	4.85	4.47	3.76	3.24	3.83	4.28	6.61	5.91	(7)
(8)		HDD10.0	4477	981	766	567	254	76	6	0	1	26	232	617	954	(8)
(9)		HDD18.3	6820	1239	999	825	499	285	111	42	54	199	488	867	1212	(9)
(10)		CDD10.0	804	0	0	0	4	51	167	268	234	78	3	0	0	(10)
(11)		CDD18.3	107	0	0	0	0	2	22	52	29	2	0	0	0	(11)
(12)		CDH23.3	923	0	0	0	1	44	253	418	192	16	1	0	0	(12)
(13)		CDH26.7	238	0	0	0	0	10	68	123	37	1	0	0	0	(13)
(14)	Wind	WSAvg	1.1	0.8	1.0	1.4	1.5	1.4	1.0	0.8	0.9	1.0	1.1	1.1	0.9	(14)
(15)	Precipitation	PrecAvg	661	18	13	22	41	79	79	108	103	84	61	28	25	(15)
(16)		PrecMax	845	42	51	65	114	145	158	204	222	181	196	75	80	(16)
(17)		PrecMin	368	1	1	1	6	14	21	3	39	9	14	1	1	(17)
(18)		PrecStd	130	13	11	18	25	35	32	56	50	45	42	22	22	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-4.7	-2.1	7.4	20.2	28.1	31.4	33.2	30.1	25.2	18.0	5.9	-2.4	(19)
(20)			MCWB	-6.0	-4.8	1.5	9.9	15.2	20.4	23.1	21.3	16.9	9.8	1.1	-3.3	(20)
(21)		2%	DB	-7.1	-5.4	4.0	15.6	23.7	28.6	29.6	27.4	22.6	13.6	2.2	-5.5	(21)
(22)			MCWB	-8.2	-7.4	-0.6	7.5	13.1	18.5	21.4	20.1	15.6	7.4	-1.2	-6.8	(22)
(23)		5%	DB	-9.9	-7.9	1.4	12.3	21.1	26.2	27.6	25.4	20.4	11.0	0.1	-8.9	(23)
(24)			MCWB	-10.9	-9.5	-2.3	5.3	11.8	17.4	20.4	19.2	14.3	5.5	-2.1	-9.9	(24)
(25)		10%	DB	-12.7	-9.9	-0.7	9.3	18.4	24.0	25.5	23.5	18.1	8.8	-1.5	-12.2	(25)
(26)			MCWB	-13.6	-11.4	-3.8	3.3	10.5	16.6	19.7	18.2	13.1	4.4	-3.5	-13.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-5.6	-4.8	1.7	10.4	16.0	22.3	24.6	23.1	18.3	10.3	1.5	-3.4	(27)
(28)			MCDWB	-4.6	-2.1	6.7	18.6	24.5	29.0	30.2	27.6	23.0	17.2	5.0	-2.6	(28)
(29)		2%	WB	-8.2	-7.4	-0.4	7.7	14.0	20.0	22.6	21.4	16.8	8.1	-0.5	-6.8	(29)
(30)			MCDWB	-7.1	-5.4	3.5	15.5	22.4	26.1	28.0	25.6	21.1	12.4	1.4	-5.8	(30)
(31)		5%	WB	-10.9	-9.4	-2.1	5.4	12.3	18.6	21.3	20.2	15.2	6.5	-2.0	-10.0	(31)
(32)			MCDWB	-10.0	-7.8	1.1	11.9	20.0	24.3	26.3	24.0	18.7	10.3	0.1	-8.9	(32)
(33)		10%	WB	-13.6	-11.3	-3.6	3.5	10.8	17.2	20.1	19.1	13.9	4.9	-3.5	-13.1	(33)
(34)			MCDWB	-12.7	-10.0	-0.9	8.8	17.3	22.6	24.5	22.5	17.0	8.2	-1.6	-12.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.9	8.3	9.2	9.1	10.9	11.2	9.5	9.3	10.2	8.1	7.0	6.7	(35)
(36)			MCDBR	7.6	9.1	10.9	13.9	16.0	15.1	13.5	12.8	13.4	11.7	7.6	7.3	(36)
(37)			MCWBR	6.8	7.4	7.5	8.3	8.1	7.3	6.6	6.4	7.0	7.0	5.5	6.7	(37)
(38)		5% WB	MCDWB	7.2	9.0	10.3	13.6	15.1	13.4	11.9	10.9	11.8	10.7	6.9	7.3	(38)
(39)			MCWBR	6.5	7.4	7.1	8.3	7.9	7.2	6.6	6.2	6.8	6.8	5.2	6.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.256	0.287	0.353	0.460	0.536	0.487	0.490	0.415	0.365	0.363	0.297	0.254	(40)	
(41)		taud	2.139	2.088	1.940	1.818	1.812	2.059	2.098	2.325	2.398	2.187	2.112	2.101	(41)	
(42)		Ebn at Noon	734	813	813	763	726	774	764	810	814	710	655	653	(42)	
(43)		Edh at Noon	70	101	146	188	199	158	149	113	93	93	73	61	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.41	2.59	4.19	5.23	4.95	5.35	4.86	4.21	3.28	2.06	1.38	1.07	(44)	
(45)		RadStd	0.05	0.08	0.18	0.34	0.54	0.56	0.49	0.43	0.31	0.16	0.10	0.06	(45)	

Historical Trends

		DBAvg	Heating		Cooling		Degree-Days			
			99% DB	99% DP	1% DB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+0.39	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page